Varad Rane | *.NET / Backend Developer*

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Summary

.NET engineer with 2+ years of production C#/.NET experience at one of India's largest brokerages, delivering trading and client-facing systems under audit and compliance requirements. Built token-based authentication for trading APIs serving 2M users, an internal API gateway, and a .NET Standard 2.1 shared library adopted by 50+ engineers across 4 teams. Migrated a legacy .NET Framework CRM to .NET 7 with zero downtime. Full-stack across ASP.NET Core, Angular and SQL Server, with deep T-SQL performance tuning.

Experience

Software Engineer II Jun 2022 - Aug 2024
Kotak Securities Mumbai, India

- Built token-based authentication from scratch in ASP.NET Core for user-facing trading APIs serving 2M users at ~260 req/s peak, 30% above the previous 200 req/s, establishing the security and audit-compliance baseline adopted by all later internal applications.
- Built a cross-compatible **.NET Standard 2.1** shared library for data access and logging, adopted by 50+ engineers across 4 teams; eliminated two network hops per request and resolved a production **IIS socket-exhaustion** failure degrading service reliability.
- Migrated a legacy CRM of 70 views from **.NET Framework to .NET 7** with zero production downtime, resolving breaking changes across the dependency graph and realigning authentication flows to updated security standards.
- Designed an internal API gateway centralizing authentication, CORS and rate limiting, with SQL-driven routing configuration that enabled parallel live-environment testing against production data without a separate staging deployment.
- Standardized authentication across backend services by integrating third-party 2FA into an internal IAM layer; partnered with external penetration testers to remediate 12 findings and authored the resulting security compliance documentation.
- Optimized high-frequency production **T-SQL** via execution-plan analysis and indexing, cutting p95 latency on critical data-retrieval paths from 237 ms to 145 ms.
- Built a reusable bulk file-upload module in ASP.NET Core with validation, transactional error handling and user-facing recovery, standardizing CSV ingestion across 7 teams and halving development time for new upload workflows.
- Architected an alert-management system across 100+ production servers and message queues with configurable health checks, cutting quarterly downtime from 6 hours to 5 and automating post-incident root-cause reporting.
- Delivered full-stack features across **Angular and ASP.NET Core** with SQL Server-backed REST APIs for internal trading and client-facing workflows.

Software Developer Feb 2026 - Present
Field Worker Princeton, NJ

- Built a document-extraction pipeline turning multi-column, cross-page PDF reports into schema-validated JSON, automating ~95% of client intake; delivered PostgreSQL storage, backend APIs and a React interface end to end as sole engineer reporting to the CTO.
- Took a local RAG prototype to production on Azure: an agentic case-summary assistant inside the Node.js API (Azure AI Foundry GPT-4.1, PostgreSQL/pgvector) with 10 tool-calling functions for per-patient summaries, risks, open issues, urgent flags and caseload-wide reviews, plus its React chat UI.
- Designed a reproducible benchmarking harness measuring latency, throughput and retrieval quality across configuration strategies, raising retrieval accuracy from 72% to 93%.

Teaching Assistant Sep 2025 - Dec 2025
New Jersey Institute of Technology Newark, NJ

- Built an automated grading system running student C++ and Python submissions against test suites in isolated containers; cut turnaround for 50 students from 5 days to 6 hours.

Software Engineering Intern Jan 2021 - Aug 2021
Renaissance Cell, VESIT Mumbai, India

- Built a full-stack certificate-distribution platform and Discord bot with college-email authentication, bulk upload of digitally signed certificates and real-time delivery via the Discord API, cutting issuance turnaround from 30 days to 10 minutes for 2,500+ students.

Projects

Cloud Document Extraction Pipeline

https://github.com/varadrane7/cloud-extraction-pipeline

Python, AWS Lambda, Step Functions, Bedrock, Aurora + pgvector, Terraform

- Serverless event-driven pipeline: S3 ingest, Step Functions Distributed Map across 8 Lambdas, Bedrock (Claude 3.5 Sonnet vision + Titan Embeddings), Aurora + pgvector with HNSW index. Modular Terraform with separate dev/prod environments; GitHub Actions deploy via OIDC federation (no static credentials) with terraform-validate gated on every PR; per-document cost tracking via CloudWatch EMF.

PrivGuard Sentinel

https://blog.varadrane.com/posts/privguard-sentinel

Python, GitHub Actions, LLM APIs

- LLM pull-request reviewer posting inline comments from structured JSON output, with diff chunking and configurable severity thresholds.

LazyCurl

https://blog.varadrane.com/posts/lazycurl

Go, Bubble Tea

- Terminal UI for HTTP APIs with session management, visual request editing and a concurrent load-testing engine.

Education

New Jersey Institute of Technology Newark, NJ

MS, Computer Science - GPA 3.90 / 4.00

University of Mumbai Mumbai, India

BE, Electronics Engineering - GPA 9.32 / 10

Skills

.NET: C#, ASP.NET Core, .NET 7, .NET Core, .NET Standard 2.1, .NET Framework, Entity Framework, LINQ, async/await, dependency injection, middleware, NuGet packaging, IIS, Web API, framework migration
Databases: Microsoft SQL Server, T-SQL, stored procedures, execution-plan analysis, indexing, query optimization, schema migration, PostgreSQL, data modeling
Backend & Architecture: REST API design, API gateways, microservices, distributed system design, OAuth 2.0, JWT, 2FA, IAM, rate limiting, transactional error handling, system design
Frontend: Angular, TypeScript, React, Next.js, JavaScript
Cloud & DevOps: AWS (Lambda, Step Functions, Bedrock, Aurora, S3), Terraform, Docker, GitHub Actions, CI/CD, Linux, monitoring and alerting
Other Languages: Python, Go, SQL, C++
Practices: Agile/Scrum, code review, unit & integration testing, performance profiling, technical and compliance documentation

Certificates and Publications

AWS Building Production-Ready AI Agents with Amazon Bedrock AgentCore (2026)

Rane, V., et al. "Automated Crack Severity Level Detection and Classification for Surface Crack Using Deep Convolutional Neural Networks." Springer: *Advanced Computing and Intelligent Technologies*, Aug 2022